MUKUL PANDEY

Full-Stack Software Engineer | Applied AI & LLM Integration

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Profile

Full-Stack Software Engineer at Volkswagen Group Digital Solutions India with hands-on experience building enterprise applications using **Java, Quarkus, REST APIs, Hibernate/JPA, SQL**, and **Vue.js**. Experienced in backend service development, database integration, API security, automated testing, and microservices maintenance. Builds and deploys full-stack applications using **React, TypeScript, Spring Boot,** and **PostgreSQL**, including AI-powered workflows integrating the **Gemini API** for intelligent issue-draft generation. Combines practical software engineering experience with an academic foundation in machine learning and deep learning.

Skills

Programming Languages: — Java, Python, JavaScript, TypeScript, SQL, C/C++

Backend & Web Development: — Spring Boot, Quarkus, Node.js, Hibernate ORM/JPA, Panache, REST APIs, API Design, React.js, Vue.js, HTML5, CSS3

AI & Machine Learning: — Gemini API, LLM/API Integration, Prompt Engineering, Structured AI Responses, AI-Assisted Workflow Automation, CNN, EfficientNetB0, Logistic Regression, Support Vector Machine (SVM), Hyperparameter Tuning

Databases & Reporting: — PostgreSQL, MySQL, Oracle, MongoDB, Power BI, Power BI Report Embedding, SQL Querying, Data Visualization

Testing & Automation: — Playwright, JavaScript/TypeScript Test Automation, End-to-End Testing, UI Automation, API Testing, JUnit 5, Mockito, Unit Testing, Integration Testing using Docker

Cloud & DevOps: — Docker, OpenShift, Argo CD, Linux, Bash/Shell Scripting, Git, GitHub, Vercel, Render, Neon

Experience

Software Engineer, *Volkswagen Group of Digital Solutions India* Oct 2024 – Present | Pune, India

- Develop and maintain Java-based enterprise microservices for the **FAP Data Warehouse** ecosystem using Quarkus, Hibernate ORM/Panache, REST APIs, and SQL, integrating backend services with Vue.js frontend applications across Jira, KPM, KPMS Halle, and KPMS Feld/NP domains.
- Develop and enhance data retrieval and reporting capabilities, including random-access lazy loading for large datasets and an export service supporting configurable column selection, asynchronous report processing, and downloadable reports.
- Develop an image service that compresses images, stores them in Amazon S3, and provides links for viewing and downloading.
- Implement and maintain scheduled jobs for Jira data ingestion and periodic cleanup of orphaned tickets in the data warehouse.
- Implement authentication and authorization using OpenID Connect (OIDC), role-based access control, and API Helper components to secure service endpoints and support authenticated access.
- Manage environment-specific configurations and deployment workflows using Helm charts, Argo CD, and OpenShift; contribute to dependency upgrades, troubleshooting, and service reliability.
- Develop and maintain unit and integration tests using JUnit 5 and Mockito, and contribute to code quality through SonarQube analysis, test coverage reporting, and CI/CD quality checks.
- Recognized through VWGDS awards for contributions to the API Helper and Export Service initiatives, including the TrendSetter, Innovation, and InstaPat awards.

Projects

DevTrack — AI-Powered Project and Issue Management Application Sep 2026

Technologies: React, TypeScript, Java 21, Spring Boot, PostgreSQL, Gemini API, REST APIs, Vercel, Render, Neon

- Designed, developed, and deployed a full-stack application for managing projects, issues, members, labels, assignees, comments, and issue-status history.
- Developed REST APIs using Spring Boot for authentication, project management, issue tracking, and related workflows, backed by PostgreSQL.
- Built a responsive frontend using React and TypeScript and integrated it with the backend APIs.
- Integrated the Gemini API to power an AI Issue Assistant that transforms rough issue descriptions into structured drafts containing a title, detailed description, acceptance criteria, and suggested labels.
- Designed the AI-assisted workflow to let users review and edit generated content before explicitly creating an issue, keeping AI generation separate from database persistence.
- Deployed the frontend on Vercel, backend on Render, and database on Neon.

Deep Learning based Disease Detection System for Plants Jan 2024 – Jun 2024

Technologies: Python, CNN, EfficientNetB0, Streamlit, hyperparameter tuning

- Developed a deep-learning-based plant disease detection application using CNN-based image classification and EfficientNetB0.
- Applied parameter tuning to improve model performance and built an interactive application using Streamlit.
- Reported 99.86% accuracy in the project evaluation; clarify the evaluation dataset and methodology when presenting this result.

Education

Master of Business Administration Jan 2025 – Present | Pune, India

Indira Gandhi National Open University, Regional Centre Pune

B.Tech CSE, *Institute of Engineering and Rural Technology Prayagraj* Nov 2020 – Jul 2024 | Prayagraj, India

I-DIV with distinction. CGPA=8.19